
Integrable Systems — Exercise Sheet 3

Symplectic manifolds

Variant: Questions only

Conventions throughout: $\omega = \sum_i dq_i \wedge dp_i$ in that order, $\iota_{X_f} \omega = df$, and $\{f, g\} = \omega(X_f, X_g)$. On one degree of freedom, $(a dx + b dy) \wedge (c dx + d dy) = (ad - bc) dx \wedge dy$, which is all the wedge algebra most problems need.

Exercise 1 (the oscillator in new coordinates). Let $\nu > 0$ and, on $\mathbb{R}^2$ minus the origin, introduce (φ, I) with $I > 0$ by

$$q = \sqrt{2I/\nu} \sin \varphi, \quad p = \sqrt{2I\nu} \cos \varphi.$$

(i) Compute $\omega = dq \wedge dp$ in the new coordinates, and say which of φ, I plays the role of a position. (ii) Express $H = \frac{1}{2}(p^2 + \nu^2 q^2)$ in the new coordinates and write down Hamilton's equations there. Solve them. (iii) Compute $\oint p dq$ around the circle $I = \text{const}$, traversed in the direction of increasing φ . Is that the direction of the motion? (iv) Repeat (i) for the “textbook” choice $q = \sqrt{2I/\nu} \cos \varphi$, $p = \sqrt{2I\nu} \sin \varphi$, and explain what goes wrong.

Exercise 2 (from form to bracket and back). On the half-plane $U = \{(x, y) : x > 0\}$ let $\omega = x^{-1} dx \wedge dy$. (i) Show that (U, ω) is a symplectic manifold. (ii) Solve $\iota_{X_f} \omega = df$ for X_f , and find $\{f, g\}$ and $\{x, y\}$. (iii) Conversely, suppose you are told only that $\{x, y\} = x$. Recover ω , using the relation $J = -W^{-1}$ of Lecture 3, where $J^{ab} = \{z^a, z^b\}$ and $W_{ab} = \omega(\partial_a, \partial_b)$. (iv) Find Darboux coordinates (q, p) on U , and describe the flow of $H = y$ in both coordinate systems.

Exercise 3 (Hamiltonian flows are symplectic). (i) Write down the time- t flow of $H = \frac{1}{2}(p^2 + \nu^2 q^2)$ and verify directly that it satisfies $dq(t) \wedge dp(t) = dq \wedge dp$. (ii) For an arbitrary smooth H on $\mathbb{R}^2$, let $D(t)$ be the Jacobian of the flow at a point, so that $\dot{D} = AD$, $D(0) = \mathbb{1}$, with $A = \Omega S$ and S the Hessian of H along the trajectory (Lecture 3, §3.4). Using $\frac{d}{dt} \det D = \text{tr}(A) \det D$, show that $\det D = 1$ for every t at which the flow is defined at that point, and conclude that the flow is symplectic wherever it is defined. Why can one not say “for all $t \in \mathbb{R}$ ”? (iii) Show by example that for $n = 2$ the condition $\det D = 1$ is *not* enough for a linear map of $\mathbb{R}^4$ to be symplectic.

Exercise 4 (a generating function). Let $F(q, Q) = \frac{\nu}{2} q^2 \cot Q$, a type-1 generating function, with $p = \partial F / \partial q$ and $P = -\partial F / \partial Q$, defined for $0 < Q < \pi$ and $q \neq 0$. (i) Find p and P , and solve for (q, p) as functions of (Q, P) . There are two branches; take the one with $q > 0$. (ii) Verify directly that the transformation preserves ω . (iii) Express $H = \frac{1}{2}(p^2 + \nu^2 q^2)$ in terms of (Q, P) and solve the motion. Compare with Exercise 1. (iv) Where does this generating function fail, and why is that not a defect of the transformation?

Exercise 5 (the Liouville measure on $\mathbb{R}^4$). (i) With $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$, compute $\omega \wedge \omega$, and hence $\omega^2/2!$, and identify it with the phase-space volume element of Lecture 1. (ii) For $F_i = \frac{1}{2}(q_i^2 + p_i^2)$ and $c_1, c_2 > 0$, compute the Liouville volume of $\{F_1 \leq c_1, F_2 \leq c_2\}$ in two ways: directly, and in the coordinates of Exercise 1 (with $\nu = 1$) applied in each plane.

Exercise 6 (three candidates). In $\mathbb{R}^4$ with coordinates (q_1, q_2, p_1, p_2) , decide which of the following are Lagrangian, and for each that is a graph of dS over q -space, find S . (a)

The plane $p_1 = q_1 + q_2$, $p_2 = q_1 - q_2$. (b) The plane $p_1 = q_2$, $p_2 = 0$. (c) The cylinder $\{q_1^2 + p_1^2 = 1, q_2 = 0\}$.

Exercise 7 (Darboux where it is elementary). (i) Let $\omega = f(x, y) dx \wedge dy$ on $\mathbb{R}^2$ with $f > 0$ everywhere. Show that $q = x$, $p = \int_0^y f(x, s) ds$ is a Darboux chart, i.e. a diffeomorphism onto its image with $dq \wedge dp = \omega$. (ii) What changes if $f < 0$ everywhere? What if f vanishes on the line $y = 0$? (iii) Explain why no such argument can make the round metric of a sphere look like $dx^2 + dy^2$ near a point, and what that contrast says.

Exercise 8 (cotangent bundles are symplectic). (i) Let $q' = \phi(q)$ be a change of coordinates on configuration space, with momenta transforming as $p_i = \sum_j p'_j \partial q'_j / \partial q_i$. Show that $\sum_j p'_j dq'_j = \sum_i p_i dq_i$, and deduce that $-d(p \cdot dq)$ is the same symplectic form in both charts. (ii) Check it in polar coordinates: with $p_r = (xp_x + yp_y)/r$ and $p_\theta = xp_y - yp_x$, show that $dx \wedge dp_x + dy \wedge dp_y = dr \wedge dp_r + d\theta \wedge dp_\theta$ without computing a single wedge product. (iii) By contrast, on $T^*\mathbb{R}$ take $q' = q^3$ (on $q > 0$). Compute $q' dp'$ and show that it is *not* $q dp$. Why is this not a contradiction with $d(q dp) = \omega$?

Exercise 9 ★. On $\mathbb{R}^4$ with coordinates (q_1, q_2, p_1, p_2) let a be a smooth, everywhere positive function of all four coordinates, and

$$\omega_a = a dq_1 \wedge dp_1 + dq_2 \wedge dp_2.$$

(i) Show that ω_a is non-degenerate, compute $d\omega_a$, and find exactly which functions a make ω_a symplectic. (ii) Compute the four fields $X_{q_1}, X_{q_2}, X_{p_1}, X_{p_2}$ and the brackets of the coordinates. (iii) Compute the Jacobi sums for the triples (q_1, p_1, p_2) and (q_1, p_1, q_2) , and check each against $d\omega_a$ evaluated on the three fields, as Lecture 3's theorem on Jacobi and $d\omega = 0$ requires. Explain, from the form of $d\omega_a$, why the Jacobi sums never involve $\partial_{q_1} a$ or $\partial_{p_1} a$. (iv) When ω_a is symplectic, find Darboux coordinates.